An explicit upper bound for reciprocal sums of sets with large pairwise least common multiples

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Abstract

For a positive integer n let $\rho(n)$ be the maximum of $\sum_{a\in A} 1/a$ over all sets $A \subseteq \{1,\dots,n\}$ such that $\operatorname{lcm}(a,b) > n$ for all distinct $a,b \in A$. We prove that $\rho(n) \le 1.0087578638 + 87.5686331624/n$ for every $n \ge 1$. In particular $\rho(n) \le 1.0170$ for $n \ge 10625$ and $\rho(n) \le 1.01$ for $n \ge 70499$. As $n \to \infty$ this improves the bound of Chen (1996), whose leading constant is 1.0170166; that bound is the strongest we located in a literature search of 12 October 2026. For $70 \le n \le 8871$ Chen's explicit bound remains smaller. The proof uses weak linear-programming duality, with dual weights that depend on m/n and on $\gcd(m,6)$. A certificate of 67 dyadic rationals reduces dual feasibility, for all n at once, to 239 980 finite inequalities and four tail inequalities. A short Python program checks all of them in exact integer and rational arithmetic. The conjecture $\rho(n) \le 1 + o(1)$ remains open.

1 Introduction

Let n be a positive integer. We call a set $A \subseteq \{1,\dots,n\}$ *lcm-separated* if $\operatorname{lcm}(a,b) > n$ for all distinct $a,b \in A$, and we put

$$\rho(n) = \max\Big\{\sum_{a\in A}\frac{1}{a} :\ A \subseteq \{1,\dots,n\} \text{ lcm-separated}\Big\}.$$

Every singleton is lcm-separated, so $\rho(n) \ge 1$; the set $\{1\}$ is allowed. Erdős asked whether $\rho(n) \le 31/30$ for all n. This is part of Problem 542 in Bloom's list [1]; Section 2 recalls what is known. Our main result is an explicit bound with leading constant below 1.0088.

Theorem 1.1. *Let V_1 and E_1 be the constants defined in* (5) *from the certificate in Appendix A. They satisfy*

$$E_1 = \frac{564156623381}{6442450944} = 87.568633162261297305\dots,$$
$$1.008757863713504319 \le V_1 \le 1.008757863713504320.$$

For every integer $n \ge 1$ and every set $A \subseteq \{1,\dots,n\}$ with $\operatorname{lcm}(a,b) > n$ for all distinct $a,b \in A$,

$$\sum_{a\in A}\frac{1}{a} \ \le\ V_1 + \frac{E_1}{n} \ \le\ 1.0087578638 + \frac{87.5686331624}{n}.$$

In other words, $\rho(n) \le 1.0087578638 + 87.5686331624/n$ for all $n \ge 1$.

Corollary 1.2. $\rho(n) \le 1.0170$ *for every $n \ge 10625$, and $\rho(n) \le 1.01$ for every $n \ge 70499$. These are the least onsets that follow from the displayed constants* 1.0087578638 *and* 87.5686331624.

The proof (Sections 3 and 4) is weak duality for the linear-programming relaxation of the problem. If A is lcm-separated, the sets of multiples $\{m \le n : a \mid m\}$, $a \in A$, are pairwise disjoint. So any nonnegative weights $z_1,\dots,z_n$ that put mass at least $1/a$ on the multiples of each $a \le n$ give $\rho(n) \le \sum_m z_m$. We take $z_m = c_{\gcd(m,6)}(m/n)/m$, where c_1, c_2, c_3, c_6 are step functions on $(1/10, 1]$ with 54 steps. Because 6 is squarefree, the dual constraints for all n at once reduce to inequalities $F_\gamma(x) \ge 1$, for real $x \in (0,1]$ and $\gamma \mid 6$, that do not involve n. The functions F_γ are piecewise constant. This reduces the range $x \ge 10^{-4}$ to

59 995 points, and an elementary estimate for harmonic sums in arithmetic progressions handles $x < 10^{-4}$. All resulting inequalities are checked in exact arithmetic.

The dependence on $\gcd(m, 6)$ lets the dual put more weight on integers with more small divisors. In floating-point experiments on the grids we tried, weights depending on m/n alone did not get below about 1.0172. This observation guided the search; it is not a statement about that family of weights, and it is not used in the proof.

Theorem 1.1 gives $\limsup_{n\to\infty} \rho(n) \le 1.0087578638$. It does not prove the conjecture $\rho(n) \le 1 + o(1)$ [1], and we do not claim that the constant is sharp (Remark 6.1). The error term $87.5686331624/n$ is explicit; no unspecified constants occur. The error term is larger than Chen's, so Theorem 1.1 does not improve Chen's explicit bound for every n (Remark 6.2).

2 Related work

Problem 542 in Bloom's list [1] asks whether every set $A \subseteq \{1, \dots, n\}$ with $\operatorname{lcm}(a, b) > n$ for all distinct $a, b \in A$ satisfies $\sum_{a\in A} 1/a \le 31/30$. It also asks a separate question about the integers $b \le n$ that divide no element of A. The problem page records both questions as solved. The constant $31/30$ cannot be lowered, since $\{2, 3, 5\}$ is lcm-separated for $n = 5$. The literature records $\rho(5) = 31/30$ and $\rho(11) = 4699/4620 = 1.01709\ldots$, the latter attained by $\{3, 4, 5, 7, 11\}$ [2, 4].

The problem page and its discussion thread describe $\rho(n) \le 1 + o(1)$ as conjectural. They attribute to Erdős, Schinzel and Szekeres the stronger conjecture that $\{2, 3, 5\}$ and $\{3, 4, 5, 7, 11\}$ are the only such sets with reciprocal sum greater than 1 [1]. The set $\{2, 3, 5\}$ is lcm-separated only for $n = 5$, and $\{3, 4, 5, 7, 11\}$ only for $n = 11$. Since $\{1\}$ is always admissible, this conjecture would give $\rho(n) = 1$ for every n other than 5 and 11, in particular for every $n \ge 12$.

Chen [2, Theorem 2, p. 104] proved that $\rho(n) < 1.0170166$ for all sufficiently large n, and explicitly

$$\rho(n) < 1.0170166 + \frac{14.21}{n - 58} \qquad \text{for every integer } n \ge 59. \tag{1}$$

The same paper notes that $\rho(n) < 1.017099 < \rho(11)$ for $n \ge 172509$. Its Theorem 1 is a conditional route to $\rho(n) \to 1$, not an unconditional result. Weighted counting over the disjoint sets of multiples already appears in [2] (Lemma 1 on p. 105, and the proof of Theorem 1). The duality principle of Lemma 3.1 is therefore not new. The contribution of this note is the family of weights, which depend on $\gcd(m, 6)$, and the exact certificate.

Chen [3, Theorem 1] also studied a variant $\rho_m(n)$ in which the elements of A are at most m while their pairwise least common multiples exceed n. For fixed $0 < \delta < 1 - \log 2$ and every $\varepsilon > 0$ he proved $|\rho_m(n) - 1| < \varepsilon$ for all sufficiently large n and $n \exp(-(\log n)^\delta) \le m \le \varepsilon n$. This regime, $m = o(n)$, does not include $m = n$. Weingartner [4, Theorem 3 and Question 1] considers a variant $R(x)$ in which the element 1 is excluded. He proves the lower bound $R(x) \ge 1 - \kappa/\log x + O((\log x)^{-3/2})$ with $0.543595 < \kappa < 0.543804$, and asks whether this is the asymptotic value. He does not prove an upper bound of the form $1 + o(1)$.

We carried out a bounded literature search on 12 October 2026. It covered the problem page with its comments and discussion thread, arXiv and targeted web searches (on the numerical constants, on primitive and lcm-separated sets, and on linear-programming and flow certificates), the OpenAlex index and the formal-conjectures repository. We also examined results on primitive sets with the weight $1/(a \log a)$ and certificate frameworks for related problems; none of them gives a bound for $\rho(n)$ of the present kind. We did not locate a bound for $\rho(n)$ with leading constant below 1.0170166. Theorem 1.1 therefore improves the strongest bound we located in this search. The search was not exhaustive: one further citation index refused our queries (rate limit) and could not be used, the OpenAlex index did not resolve the record of [2], so its forward citations could not be traced there, and the community chat archive was searched only in part.

3 Duality and the dual weights

Lemma 3.1 (weak duality)**.** *Let $n \ge 1$ and let $z_1, \dots, z_n \ge 0$ satisfy*

$$\sum_{1\le k\le n/a} z_{ka} \;\ge\; \frac{1}{a} \qquad \textit{for every } 1 \le a \le n. \tag{2}$$

Then $\rho(n) \le \sum_{m=1}^{n} z_m$.

Proof. Let $A \subseteq \{1,\dots,n\}$ be lcm-separated and put $M(a) = \{ka : k \ge 1,\ ka \le n\}$. If $m \in M(a) \cap M(b)$ for distinct $a, b \in A$, then $\operatorname{lcm}(a,b) \mid m$, so $\operatorname{lcm}(a,b) \le m \le n$, which is impossible. Hence the sets $M(a)$, $a \in A$, are pairwise disjoint, and by (2) and $z \ge 0$,

$$\sum_{a\in A} \frac{1}{a} \le \sum_{a\in A} \sum_{m\in M(a)} z_m \le \sum_{m=1}^{n} z_m. \qquad \square$$

Remark 3.2. If A is lcm-separated, every $m \le n$ has at most one divisor in A. So the indicator vector of A is feasible for the linear program: maximize $\sum_{a\le n} y_a/a$ subject to $\sum_{a\mid m} y_a \le 1$ for $1 \le m \le n$ and $y \ge 0$. Condition (2) with $z \ge 0$ describes the feasible points of the dual program, which minimizes $\sum_m z_m$, and Lemma 3.1 is weak duality for this pair.

3.1 Weights depending on m/n and on $\gcd(m,6)$

Let

$$G = \{p/j :\ 1 \le p \le 4,\ p \le j \le 10p\}.$$

This set has 55 elements $\frac{1}{10} = g_0 < g_1 < \cdots < g_{54} = 1$, listed in Appendix A. They define the 54 bins $I_b = (g_b, g_{b+1}]$, $0 \le b \le 53$. For each divisor η of 6 and each bin b the certificate fixes a number

$$c_{\eta,b} = C_{\eta,b}/2^{30}, \qquad C_{\eta,b} \in \mathbb{Z}_{\ge 0};$$

67 of the 216 integers $C_{\eta,b}$ are nonzero (Appendix A). Define step functions on $(0,1]$ by

$$c_\eta(t) = c_{\eta,b} \quad (t \in I_b), \qquad c_\eta(t) = 0 \quad (0 < t \le \tfrac{1}{10}),$$

and, for a given $n \ge 1$,

$$z_m = \frac{c_{\gcd(m,6)}(m/n)}{m} \qquad (1 \le m \le n). \tag{3}$$

For $\gamma \mid 6$ and an integer k write $\gamma \vee k = \operatorname{lcm}(\gamma, \gcd(k,6))$, with $\gcd(0,6) = 6$. For real $0 < x \le 1$ put

$$F_\gamma(x) = \sum_{1\le k\le 1/x} \frac{c_{\gamma\vee k}(kx)}{k}. \tag{4}$$

Finally define the constants

$$V_\gamma = \frac{1}{6} \sum_{r=0}^{5} \sum_{b=0}^{53} c_{\gamma\vee r,b} \log \frac{g_{b+1}}{g_b}, \qquad E_\gamma = \sum_{r=0}^{5} \sum_{b=0}^{53} \frac{c_{\gamma\vee r,b}}{g_b}. \tag{5}$$

For $\gamma = 1$ these are the constants V_1 and E_1 of Theorem 1.1.

Lemma 3.3 (reduction to inequalities free of n)**.** *Suppose that* $F_\gamma(x) \ge 1$ *for every* $\gamma \mid 6$ *and every real* $x \in (0,1]$. *Then for every* $n \ge 1$ *the weights* (3)*, which are nonnegative because every* $C_{\eta,b} \ge 0$*, satisfy* (2)*, and hence* $\rho(n) \le \sum_{m\le n} z_m$.

Proof. Let $1 \le a \le n$, $\gamma = \gcd(a,6)$ and $x = a/n$. A prime $p \in \{2,3\}$ divides ka if and only if it divides a or k; since 6 is squarefree, $\gcd(ka,6) = \gamma \vee k$. Therefore

$$a \sum_{1\le k\le n/a} z_{ka} = \sum_{1\le k\le n/a} \frac{c_{\gamma\vee k}(ka/n)}{k} = F_\gamma(x) \ge 1.$$

Lemma 3.1 completes the proof. $\square$

The hypothesis of Lemma 3.3 does not involve n; this is why a single certificate covers all n. Section 4 verifies it and bounds $\sum_{m\le n} z_m$.

4 The certificate and its exact verification

4.1 Two elementary lemmas

Lemma 4.1 (harmonic sums in progressions)**.** *Let $L \geq 1$ and r be integers and let $0 < A < B$ be real. Then*

$$\frac{1}{L}\log\frac{B}{A+L} \;\leq \sum_{\substack{A<k\leq B\\ k\equiv r \pmod L}} \frac{1}{k} \;\leq\; \frac{1}{A}+\frac{1}{L}\log\frac{B}{A}.$$

Proof. Lower bound. If $A + L \geq B$ the left side is at most 0. Otherwise every interval $(t, t + L]$ contains exactly one member of the progression, so the least member $k_0 > A$ satisfies $k_0 \leq A + L < B$; let k_1 be the largest member $\leq B$, so that $k_1 + L > B$. The intervals $[k, k + L]$, for members k with $k_0 \leq k \leq k_1$, have disjoint interiors and cover $[A + L, B]$. Since $1/k \geq \frac{1}{L}\int_k^{k+L} dt/t$, the sum is at least $\frac{1}{L}\int_{A+L}^{B} dt/t$.

Upper bound. If the sum is empty there is nothing to prove. Otherwise its first term is $1/k_0 < 1/A$. Every later term satisfies $1/k \leq \frac{1}{L}\int_{k-L}^{k} dt/t$, and these intervals $[k - L, k]$ have disjoint interiors and lie in $[A, B]$. □

Lemma 4.2 (piecewise constancy)**.** *Let $S = \{p/N : \ 1 \leq p \leq 4, \ N \geq 1\}$. For $x \in (0, 1]$ let $v(x) = \min\{s \in S : s \geq x\}$. Then $F_\gamma(x) = F_\gamma(v(x))$ for every $\gamma \mid 6$.*

Proof. The minimum exists because $S \cap [x, \infty)$ is finite (it lies in $\{p/N : N \leq 4/x\}$) and contains 1. Write $v = v(x)$. By minimality, for every $s \in S$ we have $x \leq s$ if and only if $v \leq s$. For each $k \geq 1$, the k-th term of (4) is determined by which of the following holds: $kx \leq g_0$ (the term is 0); $g_b < kx \leq g_{b+1}$ for some $0 \leq b \leq 53$ (the term is $c_{\gamma\vee k,b}/k$); or $kx > 1$ (the term is absent). These conditions are Boolean combinations of conditions $x \leq g_i/k$, and $g_i/k = p'/(jk) \in S$ whenever $g_i = p'/j$ with $p' \leq 4$. So x and v satisfy the same conditions. □

4.2 The range $10^{-4} \leq x \leq 1$: an exact integer check

Let $x_1 = 10^{-4}$. Since $x_1 \in S$ and $1 \in S$, Lemma 4.2 maps $[x_1, 1]$ into $S \cap [x_1, 1]$. In lowest terms the elements of $S \cap [x_1, 1]$ are the fractions $v = p/N$ with $1 \leq p \leq 4$, $p \leq N \leq 10^4 p$ and $\gcd(p, N) = 1$. There are $10000 + 9999 + 19998 + 19998 = 59\,995$ of them, so it suffices to verify the $4 \cdot 59\,995 = 239\,980$ inequalities $F_\gamma(v) \geq 1$. For each of these points $\lfloor N/p \rfloor \leq 10^4$.

Integer enclosure. Put $Q = 2^{48}$ and, for $1 \leq k \leq 10^4$, $h_{\gamma,b}(k) = \lfloor 2^{18} C_{\gamma\vee k,b}/k \rfloor$, so that

$$\frac{h_{\gamma,b}(k)}{Q} \leq \frac{c_{\gamma\vee k,b}}{k} < \frac{h_{\gamma,b}(k)+1}{Q}.$$

For $v = p/N$ let $K_i = \lfloor g_i N/p \rfloor$ $(0 \leq i \leq 54)$, the number of $k \geq 1$ with $kv \leq g_i$. The k with $kv \in I_b$ are exactly those with $K_b < k \leq K_{b+1}$, the $k \leq K_0$ contribute 0, and $K_{54} = \lfloor N/p \rfloor$ is the cutoff $k \leq 1/v$. Hence

$$T_\gamma(v) := \sum_{b=0}^{53} \sum_{K_b<k\leq K_{b+1}} h_{\gamma,b}(k) \quad \text{satisfies} \quad \frac{T_\gamma(v)}{Q} \leq F_\gamma(v) < \frac{T_\gamma(v)+10^4}{Q}. \tag{6}$$

Arithmetic model. The program `verify.py` computes $T_\gamma(v)$ from the prefix sums $\sum_{k\leq K} h_{\gamma,b}(k)$ and the integers K_i, which it obtains by exact integer division. All proof arithmetic uses Python integers and `fractions.Fraction`; no floating-point number enters the proof. The prefix sums are stored for $K \leq 40\,000$ (only $K \leq 10^4$ is queried) in arrays of signed 64-bit integers. Every stored prefix is at most

$$2\,650\,000\,000 \cdot 2^{18} \cdot H_{40000} \leq 2\,650\,000\,000 \cdot 2^{18} \cdot 16 = 11\,114\,905\,600\,000\,000 < 2^{63},$$

because every $C_{\eta,b} < 2\,650\,000\,000$ and $H_{40000} = \sum_{k\leq 40000} 1/k \leq 16$; for the latter, split $1, \ldots, 40000$ into the 16 blocks $[2^j, 2^{j+1})$, $0 \leq j \leq 15$, each contributing at most 1. An out-of-range store would raise an error rather than wrap around.

Result. The minimum of $T_\gamma(v)/Q$ over all 239 980 pairs is

$$\frac{281475258209621}{2^{48}} = 1.000001000085223523\ldots > 1,$$

attained at $(v,\gamma) = (4/15, 1)$. By (6), $F_\gamma(x) \geq 1$ for all $x \in [x_1, 1]$ and all $\gamma \mid 6$.

Exact minimum. The program also evaluates $F_\gamma(v)$ exactly, as a rational number, for every pair with $T_\gamma(v)/Q \leq F_1(4/15)$. Only $(4/15, 1)$ qualifies; every other pair has $F_\gamma(v) \geq T_\gamma(v)/Q > F_1(4/15)$. So the minimum over the finite range is

$$F_1(4/15) = \frac{C_{2,48}}{2 \cdot 2^{30}} + \frac{C_{3,52}}{3 \cdot 2^{30}} = \frac{6442457387}{6442450944} = 1.0000010000852247\ldots$$

Here $k = 1, 2, 3$, with $kv = 4/15 \in I_{36}$, $8/15 \in I_{48} = (1/2, 4/7]$ and $4/5 \in I_{52} = (3/4, 4/5]$; the term $k = 1$ vanishes because $C_{1,36} = 0$.

4.3 The range $0 < x < 10^{-4}$

Let $0 < x < x_1$ and $\gamma \mid 6$. Group the terms of (4) by the residue r of k modulo 6 and by bin. We have $\gamma \vee k = \gamma \vee r$; moreover $kx \in I_b$ if and only if $g_b/x < k \leq g_{b+1}/x$, and the terms with $kx \leq g_0$ vanish. Hence

$$F_\gamma(x) = \sum_{r=0}^{5} \sum_{b=0}^{53} c_{\gamma\vee r,b} \sum_{\substack{g_b/x < k \leq g_{b+1}/x \\ k \equiv r \pmod 6}} \frac{1}{k}.$$

The lower bound of Lemma 4.1 with $L = 6$, $A = g_b/x$ and $B = g_{b+1}/x$, together with $c \geq 0$, $\log(1+y) \leq y$ for $y \geq 0$, and $E_\gamma \geq 0$, gives

$$F_\gamma(x) \geq \sum_{r,b} c_{\gamma\vee r,b}\Big(\frac{1}{6}\log\frac{g_{b+1}}{g_b} - \frac{1}{6}\log\Big(1 + \frac{6x}{g_b}\Big)\Big) \geq V_\gamma - xE_\gamma \geq V_\gamma - x_1 E_\gamma. \tag{7}$$

Rational logarithms. For rational $y > 1$ put $s = (y-1)/(y+1) \in (0,1)$ and $\ell(y) = 2\sum_{j=0}^{29} s^{2j+1}/(2j+1)$. Then

$$0 \leq \log y - \ell(y) = 2\int_0^s \frac{t^{60}}{1-t^2}\,dt \leq \frac{2s^{61}}{61(1-s^2)}.$$

Using these bounds for $\log(g_{b+1}/g_b)$ and $c \geq 0$, the program computes rationals $V_\gamma^- \leq V_\gamma \leq V_\gamma^+$; E_γ is rational. It checks $V_\gamma^- - x_1 E_\gamma - 1 > 0$ for the four values of γ in exact arithmetic. Table 1 lists the results, with decimals rounded outward.

γ	$V_\gamma \in$	E_γ (exact)	$V_\gamma^- - x_1 E_\gamma - 1 \geq$
1	[1.008757863713504319, 1.008757863713504320]	564156623381/6442450944	0.000001000397278190
2	[1.221116740952808711, 1.221116740952808712]	318232260905/3221225472	0.211237511334417068
3	[1.128746579478093456, 1.128746579478093457]	107202918283/1073741824	0.118762529691931825
6	[1.260428018694485595, 1.260428018694485596]	99593551913/1073741824	0.251152646376214048

Table 1: The tail inequalities of Section 4.3.

So $F_\gamma(x) > 1$ for $0 < x < x_1$. Together with Section 4.2, $F_\gamma \geq 1$ on $(0, 1]$ for every $\gamma \mid 6$.

4.4 Total mass; proofs of Theorem 1.1 and Corollary 1.2

Let $n \geq 1$. Group the $m \leq n$ by residue r modulo 6 and by bin: $m/n \in I_b$ if and only if $ng_b < m \leq ng_{b+1}$, and the terms with $m/n \leq g_0$ vanish. The upper bound of Lemma 4.1 with $L = 6$, $A = ng_b$ and $B = ng_{b+1}$, which also holds when no member of the progression lies in $(A, B]$, gives

$$\begin{aligned}\sum_{m=1}^{n} z_m &= \sum_{r=0}^{5}\sum_{b=0}^{53} c_{\gcd(r,6),b} \sum_{\substack{ng_b < m \leq ng_{b+1} \\ m \equiv r \pmod 6}} \frac{1}{m} \\ &\leq \sum_{r=0}^{5}\sum_{b=0}^{53} c_{\gcd(r,6),b}\Big(\frac{1}{ng_b} + \frac{1}{6}\log\frac{g_{b+1}}{g_b}\Big) = V_1 + \frac{E_1}{n},\end{aligned}$$

because $1 \vee r = \gcd(r, 6)$.

Proof of Theorem 1.1. By Sections 4.2 and 4.3 the hypothesis of Lemma 3.3 holds, so $\rho(n) \le \sum_{m \le n} z_m \le V_1 + E_1/n$ for every $n \ge 1$. In exact rational arithmetic the program verifies that $V_1^+ \le 10087578638/10^{10}$ and $E_1 \le 875686331624/10^{10}$. The stated values of V_1 and E_1 are those of Table 1. □

Proof of Corollary 1.2. The bound $1.0087578638 + 87.5686331624/n$ decreases in n. For $B \in \{1.0170, 1.01\}$ the least n with $1.0087578638 + 87.5686331624/n \le B$ is $\lceil 87.5686331624/(B - 1.0087578638) \rceil$. Since $87.5686331624/0.0082421362 = 10624.50\ldots$ and $87.5686331624/0.0012421362 = 70498.41\ldots$, these are 10625 and 70499. The program confirms both values in exact rational arithmetic. □

5 How to reproduce

The directory `code/` of the deposit contains:

- `cert_main.txt`: the certificate, printed in Appendix A; its SHA-256 is `361c296fad7662dadc53699aead4f2eb33fdbff7e0d18efd3971535acec8534e`;
- `verify.py`: the exact verifier, which alone carries the proof, and its output `verify_output.txt`;
- `crosscheck.py`: an independent check for concrete n, and its output `crosscheck_output.txt`;
- `mkcert.py`: the rounding step that produced the certificate from a floating-point solution (not part of the proof).

Certificate format. The first line, `6 4 1/10 1/10000`, gives $L = 6$, $P = 4$, $\delta = 1/10$ and x_1. Each further line $\eta\ \ b\ \ C$ gives $C_{\eta,b}$, with bins indexed $0, \ldots, 53$; pairs that are not listed are zero. The file lists 67 distinct pairs, and the verifier rejects duplicates. Appendix A prints the certificate.

Commands. Run the following from the directory containing `main.tex`; the programs need only the Python 3 standard library:

```
python3 code/verify.py code/cert_main.txt > verify.rerun.txt
diff code/verify_output.txt verify.rerun.txt
python3 code/crosscheck.py code/cert_main.txt \
  1 2 5 11 100 997 1000 5040 20011 > crosscheck.rerun.txt
cmp code/crosscheck_output.txt crosscheck.rerun.txt
```

The verifier takes about 2 CPU seconds and 72 MiB of memory and ends with `RESULT OK`; only its last line, which reports the CPU time, may differ between machines. It checks the four tail inequalities (Table 1); all 239 980 finite-range inequalities and the exact minimum (Section 4.2); the two theorem constants (Section 4.4); both onsets of Corollary 1.2; and the overlap arithmetic of Remark 6.3. As a negative control, lowering the single entry $C_{2,48}$ by 10^4 makes the verifier print `FAIL finite range` and `RESULT FAIL`, with exact minimum $6442427387/6442450944 = 0.99999634\ldots$ at $(4/15, 1)$.

The cross-check takes about 0.5 seconds and is an independent code path. For each given n it builds the weights (3) by exact rational bin lookup and tests $a \sum_k z_{ka} \ge 1$ for every $a \le n$ directly, enclosing each term by integer floor and ceiling with denominator 2^{48}. It also checks that the upper enclosure of $\sum_m z_m$ is at most $1.0087578638 + 87.5686331624/n$. Table 2 shows the results. For $n = 5$ and $n = 11$ the totals exceed $\rho(5) = 1.0333\ldots$ and $\rho(11) = 1.01709\ldots$, as they must.

How the certificate was found (not part of the proof). A floating-point linear program, solved with the HiGHS solver through SciPy, minimized V_1 over nonnegative coefficients subject to the finite-range constraints of Section 4.2 and the linear tail condition of Section 4.3. The program `mkcert.py` scaled the solution by $1 + 10^{-6}$, rounded each entry up to a multiple of 2^{-30}, and dropped entries whose scaled value was below 10^{-12}. Because this cutoff is applied after the rounding, the output is not literally an upward rounding of every entry. The LP script and its output are not included: the correctness of the certificate rests only on the exact check by `verify.py`.

n	$\min_{a\le n} a\sum_k z_{ka} \ge$	$\sum_{m\le n} z_m \le$
1	1.000001000240445	1.000001000240446
2	1.054504025727510	1.054504025727511
5	1.000001000240445	1.064258680849652
11	1.000001000240445	1.031375302518033
100	1.000001000240445	1.000001000367039
997	1.000001000085223	1.009992335549399
1000	1.000001000085223	1.009017157250768
5040	1.000001000085223	1.008592039315766
20011	1.000001000085223	1.008750262227068

Table 2: Output of `crosscheck.py` (decimals rounded outward).

6 Remarks

Remark 6.1 (small n; an observation, not a theorem). An exhaustive computer search gives $\rho(5) = 31/30$, $\rho(11) = 4699/4620$, and $\rho(n) = 1$ for every other $n \le 29$. In particular $\rho(n) = 1$ for $12 \le n \le 29$, as the conjecture of Erdős, Schinzel and Szekeres predicts. The search was run independently, with different programs, by the agent that found the certificate and by both referees (see the section on provenance). It is not part of the deposit, and nothing in this paper depends on it. In floating-point computations, the solver returned the optimal value 1.0 for the linear program of Remark 3.2 at $n = 12, \dots, 30$ and at $n = 60$, 100, 200, 400, 1000, 2000 and 5000. These computations are not proofs. If $\rho(n) = 1$ for all $n \ge 12$, then the constant 1.0087578638 in Theorem 1.1 is not sharp.

Remark 6.2 (comparison with Chen's explicit bound). Put $B(n) = 1.0087578638 + 87.5686331624/n$, and let $B_C(n) = 1.0170166 + 14.21/(n - 58)$ be the bound (1). For $n > 58$, the function $n(n - 58)(B(n) - B_C(n))$ is a quadratic polynomial in n with negative leading coefficient $1.0087578638 - 1.0170166$. It is negative at $n = 69$ and $n = 8872$, and positive at $n = 70$ and $n = 8871$. Hence $B(n) < B_C(n)$ for $59 \le n \le 69$ and for all $n \ge 8872$, while $B(n) > B_C(n)$ for $70 \le n \le 8871$. So Theorem 1.1 improves (1) for every $n \ge 8872$ (for $59 \le n \le 69$ both bounds exceed 2), but not for $70 \le n \le 8871$.

Remark 6.3 (a bound for all $n \ge 59$). Combining Corollary 1.2 with (1) gives $\rho(n) \le 1.0170 + 15/(n - 58)$ for every $n \ge 59$. For $59 \le n \le 47648$ this follows from (1), because

$$\frac{0.79}{n - 58} \ge \frac{0.79}{47590} = 0.0000166 + \frac{3}{23795000000};$$

for $n \ge 10625$ it follows from Corollary 1.2. This bound has the same shape as (1), with leading constant 1.0170 instead of 1.0170166.

Remark 6.4 (limits of the method). In (7) the tail costs $x_1 E_1 = 0.00875686\dots$ for $\gamma = 1$, almost the whole excess $V_1 - 1 = 0.00875786\dots$. Lowering the leading constant within this scheme needs a smaller x_1, which enlarges the finite check, or a sharper tail estimate that uses cancellation between adjacent bins. We have not pursued either. A proof of $\rho(n) \le 1 + o(1)$ along these lines would need dual weights whose total mass tends to 1 uniformly in n.

A The certificate

The 55 points of G in increasing order, $g_0, \dots, g_{54}$:

> 1/10, 4/39, 3/29, 2/19, 3/28, 4/37, 1/9, 4/35, 3/26, 2/17, 3/25, 4/33, 1/8, 4/31, 3/23, 2/15, 3/22, 4/29, 1/7, 4/27, 3/20, 2/13, 3/19, 4/25, 1/6, 4/23, 3/17, 2/11, 3/16, 4/21, 1/5, 4/19, 3/14, 2/9, 3/13, 4/17, 1/4, 4/15, 3/11, 2/7, 3/10, 4/13, 1/3, 4/11, 3/8, 2/5, 3/7, 4/9, 1/2, 4/7, 3/5, 2/3, 3/4, 4/5, 1.

Table 3 lists every nonzero $C_{\eta,b}$; the coefficients are $c_{\eta,b} = C_{\eta,b}/2^{30}$. Bins that are not listed have all four entries equal to zero, and an empty cell means zero. The table is generated from `code/cert_main.txt`.

b	I_b	$C_{1,b}$	$C_{2,b}$	$C_{3,b}$	$C_{6,b}$
2	(3/29, 2/19]			50643	
3	(2/19, 3/28]		20154050	50643	
5	(4/37, 1/9]		9768029		
6	(1/9, 4/35]			4658048	
9	(2/17, 3/25]		8323658		
10	(3/25, 4/33]			22116440	
14	(3/23, 2/15]			61359920	
15	(2/15, 3/22]	55180898		12973452	
17	(4/29, 1/7]		64493909		
18	(1/7, 4/27]	9910099			19444119
22	(3/19, 4/25]			45106545	
23	(4/25, 1/6]	24103401			11334220
26	(3/17, 2/11]			123858647	5705519
27	(2/11, 3/16]		31148760	63505349	
28	(3/16, 4/21]			26854466	
29	(4/21, 1/5]			26854466	
32	(3/14, 2/9]		25444426		
33	(2/9, 3/13]				5996100
34	(3/13, 4/17]			111272515	
35	(4/17, 1/4]		46095729	111272515	
36	(1/4, 4/15]		51129122	39888725	
37	(4/15, 3/11]	24420072			
38	(3/11, 2/7]			141968178	8060998
39	(2/7, 3/10]		121397974	91920816	
41	(4/13, 1/3]			85440949	
42	(1/3, 4/11]			142445358	
43	(4/11, 3/8]		4111939	208794678	
44	(3/8, 2/5]		40398029		
45	(2/5, 3/7]		98292472		
46	(3/7, 4/9]		98292472	168295214	
47	(4/9, 1/2]			198025509	
48	(1/2, 4/7]	1073742898	1153520347	1073742898	1073742898
49	(4/7, 3/5]	1073742898	1168970128	1073742898	1173837621
50	(3/5, 2/3]	1073742898	1411766076	1375392373	1375392373
51	(2/3, 3/4]	1073742898	2147485796	1522831080	2250326526
52	(3/4, 4/5]	1073742898	2199508655	1490948173	2349946970
53	(4/5, 1]	1073742898	2264530152	1643220683	2640265779

Table 3: The certificate: all 67 nonzero entries.

Provenance and verification status

The theorem, the certificate and the programs `verify.py` and `crosscheck.py` were found and written by a Claude Opus 5.5 agent (Anthropic) on 2026-10-12. The first version checked the finite range in floating point. Two isolated cross-vendor referees (GPT-6 Astra, OpenAI) reviewed it. Both found only repairable issues and asked for the same two substantive repairs: the floating-point check relied on rounding assumptions that were not justified, and one decimal value of V_1 had been printed rounded down. The certificate was kept unchanged. The verifier was rewritten in exact integer and rational arithmetic, all printed decimals are now rounded outward, and the theorem constants are compared exactly; these are the programs published here. The referees also asked that statements about floating-point LP experiments be limited to the grids actually tested, and one referee asked that the certificate conventions be stated; both requests are met here. A repair-confirmation review by one of the referees found all points of both reports addressed. It asked for one further correction, to the sentence bounding the stored prefix sums (Section 4.2), and the corrected sentence is used here. Thus both referee reports returned the verdict PASS-WITH-REPAIRS, and all requested repairs have been made. The repair confirmation also returned PASS-WITH-REPAIRS (for the one correction just described, which has not been re-reviewed); in it the referee re-verified the certificate exactly with an independent program. Each referee recomputed the finite-range check with an independent JavaScript implementation and obtained the same integer lower bound and the same exact minimizer. Both confirmed the small values of Remark 6.1 by exhaustive search. Separately, a Claude verifier agent checked the certificate in exact rational arithmetic, first with the first version of the report and again on the files of this deposit; both checks passed, and its check of the final text asked only for wording corrections, which

are made here. The literature search of Section 2 was a bounded search on 12 October 2026. The present text was prepared by the Claude agent that found the result, from the refereed report. No formal (Lean) verification has been carried out.

References

[1] T. F. Bloom, *Erdős Problems*, Problem 542, `https://www.erdosproblems.com/542`, and its discussion thread `https://www.erdosproblems.com/forum/thread/542`, accessed 12 October 2026.

[2] Y.-G. Chen, *On a problem of P. Erdős*, Acta Scientiarum Mathematicarum (Szeged) **62** (1996), 101–114. `https://acta.hu/download.phtml?id=2350`.

[3] Y.-G. Chen, *On a paper of Schinzel and Szekeres*, Acta Scientiarum Mathematicarum (Szeged) **63** (1997), 369–379. `https://acta.hu/download.phtml?id=2393`.

[4] A. Weingartner, *The Schinzel–Szekeres function*, arXiv:2310.13038 (2023; version 2, 13 June 2025). `https://arxiv.org/abs/2310.13038`.